WEBSITE TRAFFIC ANALYSIS

Alfido Tech Data Analytics Internship

Student Details

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Internship Role: Data Analytics Intern
Project Title: Website Traffic Analysis
Tools Used: Python, Pandas, NumPy, Matplotlib, Google Colab
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Introduction

Website traffic analysis is an important process for understanding user behavior, content performance, audience reach, and engagement patterns. This project aims to analyze website traffic data to identify interaction trends, evaluate user engagement levels, and generate recommendations that can help improve website performance and conversion opportunities.

Dataset Overview

The dataset contains website interaction records, including user events, geographic information, artist information, and content identifiers.

Dataset Details

- Number of Records: 226,278
- Number of Columns: 9

Key Attributes

- Event
- Date
- Country
- City
- Artist
- Album
- Track
- ISRC

- Link ID

This dataset provides valuable insights into user interactions, content engagement, and traffic distribution across different countries and cities.

Data Cleaning and Preparation

The dataset was cleaned and prepared using Python in Google Colab.

The following steps were performed:

- Loaded the dataset and examined its structure.
- Checked for missing values.
- Identified missing values in the Country, City, Artist, Album, Track, and ISRC columns.
- Replaced missing values with "Unknown".
- Converted the Date column into datetime format.
- Created a Month column for traffic analysis.
- Ensured that the dataset was suitable for analysis and visualization.

	0
event	0
date	0
country	11
city	11
artist	37
album	5
track	5
isrc	7121
linkid	0

dtype: int64

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 226278 entries, 0 to 226277
Data columns (total 9 columns):
 #   Column   Non-Null Count   Dtype
---  ------   --------------   -----
 0   event    226278 non-null  object
 1   date     226278 non-null  object
 2   country  226267 non-null  object
 3   city     226267 non-null  object
 4   artist   226241 non-null  object
 5   album    226273 non-null  object
 6   track    226273 non-null  object
 7   isrc     219157 non-null  object
 8   linkid   226278 non-null  object
dtypes: object(9)
memory usage: 15.5+ MB
```

Traffic Metrics

The following metrics were calculated to understand overall website activity:

Metric	Value

Total Events	226,278
Unique Links	3,839
Unique Countries	212
Unique Cities	11,993

These metrics provide a clear overview of website reach and user engagement.

```
total_events = len(df)
print("Total Events:", total_events)
Total Events: 226278

unique_links = df["linkid"].nunique()
print("Unique Links:", unique_links)
Unique Links: 3839

unique_countries = df["country"].nunique()
print("Unique Countries:", unique_countries)
Unique Countries: 212

unique_cities = df["city"].nunique()
print("Unique Cities:", unique_cities)
Unique Cities: 11993
```

Event Analysis

The dataset contains three primary event types:Pageviews represented the largest share of user interactions, indicating strong content visibility across the platform.

Event Type	Count
Pageview	142,015
Click	55,732
Preview	28,531

```
event
pageview    142015
click        55732
preview      28531
Name: count, dtype: int64
```

Geographic Traffic Analysis

Top Countries by Traffic

Country	Events
Saudi Arabia	47,334
India	42,992
United States	32,558
France	15,661
Iraq	8,260

Saudi Arabia generated the highest amount of website traffic, followed by India and the United States.

```
country
Saudi Arabia            47334
India                   42992
United States           32558
France                  15661
Iraq                     8260
United Kingdom           5970
Pakistan                 5644
Germany                  4794
United Arab Emirates     3702
Turkey                   3514
Name: count, dtype: int64
```

Top Cities by Traffic

City	Events
Jeddah	22,791
Riyadh	17,454
Unknown	16,672

Hyderabad	3,682
Dammam	3,276

Jeddah and Riyadh contributed the highest volume of website traffic among all cities.

```
city
Jeddah         22791
Riyadh         17454
Unknown        16672
Hyderabad       3682
Dammam          3276
Erbil           3105
Kuwait City     3010
Lucknow         2808
Muscat          2717
Delhi           2567
Name: count, dtype: int64
```

Content Engagement Analysis

Top Artists by Engagement

Artist	Events
Tesher	40,841
Anne-Marie	10,650
Tundra Beats	9,751
Surf Mesa, Emilee	7,533
DMNDS, Strange Fruits Music, Fallen Roses, Lujavo, Nito-Onna	5,512

Tesher received the highest level of engagement among all artists in the dataset.

```
artist
Tesher                                                      40841
Anne-Marie                                                  10650
Tundra Beats                                                 9751
Surf Mesa, Emilee                                            7533
DMNDS, Strange Fruits Music, Fallen Roses, Lujavo, Nito-Onna 5512
Reyanna Maria                                                5437
Shawn Mendes, Tainy                                          5409
50 Cent, Olivia                                              5367
Roddy Ricch                                                  5093
Olivia Rodrigo                                               4115
Name: count, dtype: int64
```

Data Visualizations

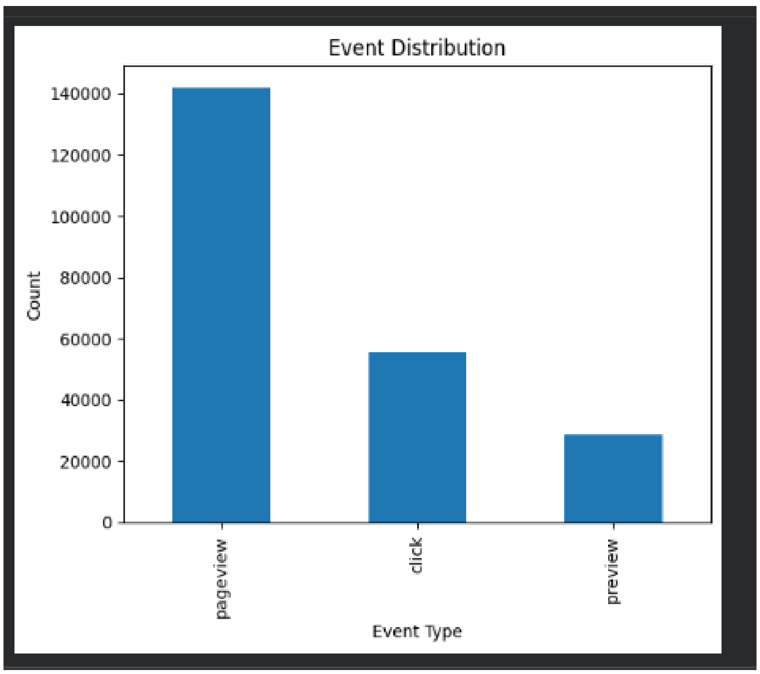

Figure 1: Event Distribution

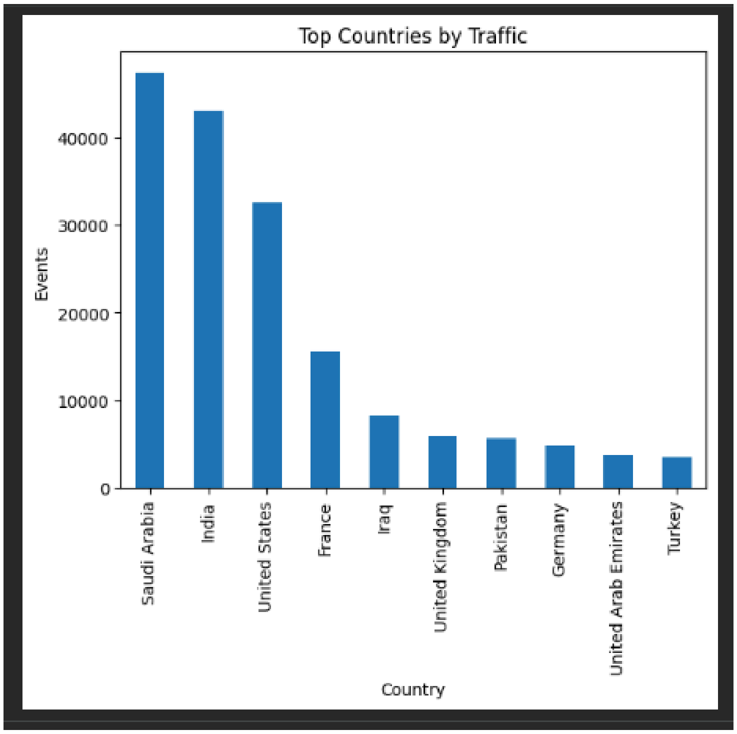

Figure 2: Top Countries by Traffic

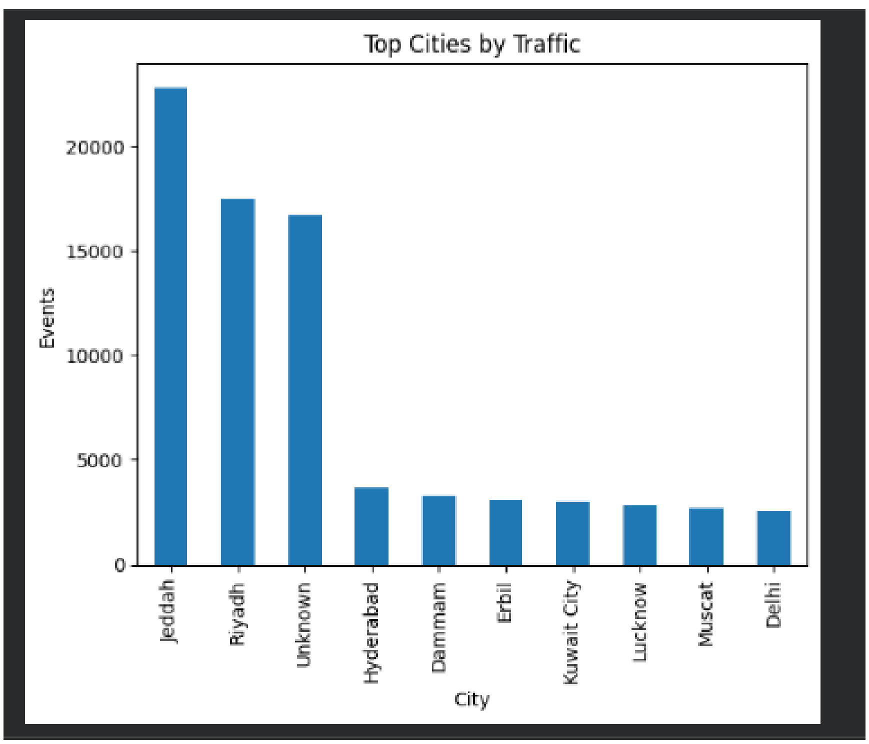

Figure 3: Top Cities by Traffic

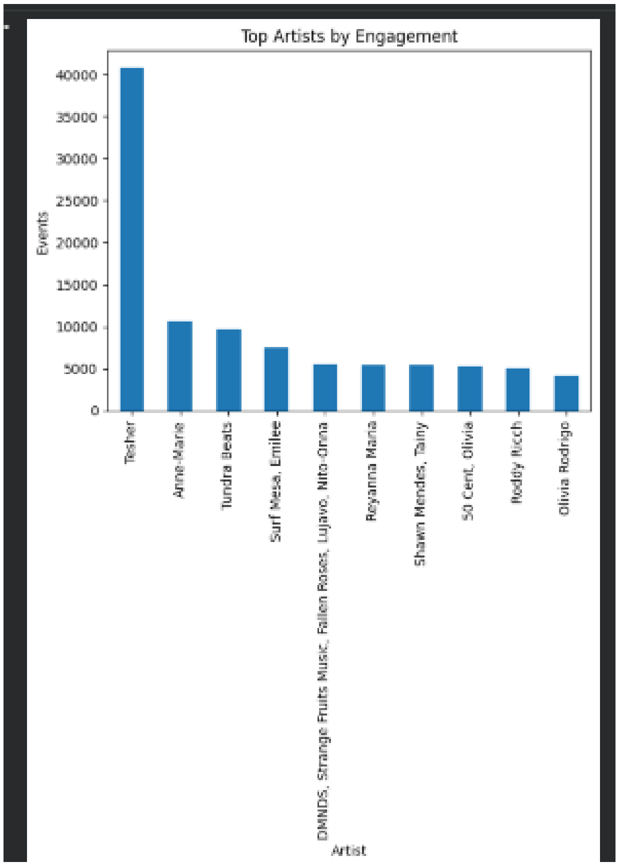

Figure 4: Top Artists by Engagement

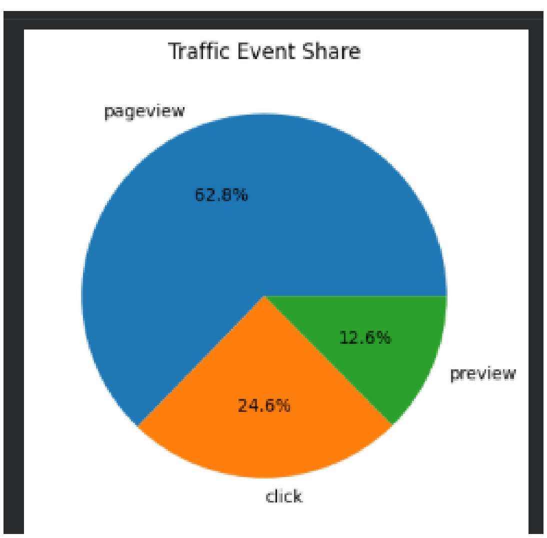

Figure 5: Traffic Event Share

Key Findings

1. A total of 226,278 user interaction events were recorded.
2. The platform attracted traffic from 212 countries and 11,993 cities.
3. Pageviews represented the largest share of interactions with 142,015 events.
4. Saudi Arabia and India were the leading traffic sources.
5. Jeddah and Riyadh generated the highest city-level traffic.
6. Tesher received the highest engagement among all artists.
7. User engagement was concentrated around a small group of highly performing content creators.
8. The dataset contained traffic records primarily from August, limiting long-term trend analysis.

Recommendations

1. Focus marketing campaigns on high-traffic countries such as Saudi Arabia, India, and the United States.
2. Develop localized content and promotional strategies for top-performing cities.
3. Highlight high-performing artists and content through featured placements and recommendations.
4. Encourage users who only view content to take additional actions such as clicks and conversions.
5. Improve tracking and data collection processes to reduce unknown geographic records and improve analytical accuracy.

Conclusion

The website traffic analysis provided valuable insights into user engagement, geographic reach, and content performance. The findings indicate that website traffic is concentrated in specific regions and around a limited number of highly engaging content creators. By focusing on high-performing markets, optimizing content recommendations, and improving user engagement strategies, organizations can enhance traffic quality and increase conversion opportunities.

EXECUTIVE SUMMARY

This project analyzed website traffic data to understand user engagement, geographic reach, and content performance. A total of 226,278 user interaction events were examined across 212 countries and 11,993 cities. Pageviews represented the majority of website interactions, while Saudi Arabia, India, and the United States were identified as the leading traffic sources. Content engagement analysis showed that a small number of artists generated a significant share of user interactions. Based on these findings, recommendations were provided to improve localized marketing efforts, promote high-performing content, enhance user engagement, and increase conversion opportunities through targeted optimization strategies.